

Chan Yat Kwan

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EDUCATION

The University of Hong Kong | Bachelor of Engineering in Computer Science | CGPA: 3.65 **June 2026**

Relevant Coursework: Data Structures and Algorithms, Database, Machine Learning, Cyber Security, OOP, Game Development, Mobile App Development, World Wide Web, Communication Network, Computer Organization, Software Engineering, Data Science

WORK EXPERIENCE

XR + AI Developer Intern | VOTANIC Limited | Unity, C#, Docker, Node.js, Express.js, HLSL **June 2025 - August 2025**

Develop XR and AI solution with XR platform integration

- Developed demo project and software package in **Unity** in a team of 2, utilizing **C#** and **machine learning** library for face detection, to implement **off-axis projection** in rendering 3D environments by tracking user head movement.
- Integrated a **photo-to-3D model generation** pipeline into a web-based 3D model editing application, utilizing **Docker**, **Node.js**, and **Express.js** to containerize and host the system with pipeline integration.
- Assisted in implementing 3D splat model rendering in **Unity** using **compute shader** with **HLSL**.

STEM Intern | Up Innovations Limited | Unity, C#, AWS, Firebase

July 2024 - August 2024

Develop interactive applications for Special Educational Needs

- Researched technologies and tools (e.g. **AWS**, **Firebase**, HuggingFace API) deployable on the projects.
- Designed and built a game prototype in **Unity** and scripted in **C#**, utilizing **machine learning** framework for **facial gesture recognition**, which helps users do oral motor exercise.
- Assisted in implementing **text-to-speech** and **speech-to-text** functions in **Unity** with HuggingFace API and **AWS lambda**.

PROJECTS

[Video Game with Optimization & AI Player](#) | *Unity, C#, Firebase, PyTorch*

Worked individually to make a 2D action game with performance optimization and training of AI Player

- Designed and developed game in **Unity**, scripted in **C#**, utilized **Firebase** to provide **player authentication** and **cloud data save-load**.
- Utilized **state design pattern** to implement game character behaviors, to improve maintainability and scalability of behavior system.
- Improved the game performance by implementing **parallel processing** that increases game frame rate by 90%; and implementing **object pooling** that reduces high frame time occurrence and avoids game stuttering.
- Trained **AI models** for simulating player behavior in playing the game using **reinforcement learning** and **imitation learning**.

[Online Library Catalogue](#) | *HTML, Firebase, JavaScript, CSS*

Worked individually to make a book searching website for an organization

- Designed and built a static website with **HTML**, **JavaScript**, **CSS** that allows users to search for information and current status of books from the organization's library, with custom search criteria and sorting on search results.
- Utilized **Firebase** to provide backend service on **authorization** and **database**, that allows upload and update book records online.

[Viral Uprising](#) | *Unity, C#*

Worked in a team of 2 to make a 3D shooting & strategic game

- Collaboratively designed and built game in **Unity**, scripted in **C#**.
- Implemented dynamic enemy NPC behavior with aggro system, and movement behavior assisted with A* pathfinding.
- Implemented behavior of strategic buildings deployable by players, and implemented shop and upgrade system.

[PlanSmart](#) | *Kotlin, Android Studio, Firebase*

Worked in a team of 2 to make a mobile app for scheduling and planning

- Collaboratively designed and built mobile app in **Android Studio**, scripted in **Kotlin**.
- Implemented functions that allow users to add, edit, and view items and fields information.
- Utilized **Firebase** to implement registration of users with email accounts and data save load system, allowing user data synchronization on multiple devices.

TECHNICAL SKILLS

Language: C#, JavaScript, HTML, CSS, Python, Java, C++, Kotlin, R

Technical Framework: .NET, Node.js, Express.js

Technical Tools / Software: Unity, Git, VS Code, Docker, Firebase, AWS, Android Studio

LANGUAGE

Cantonese - Native;

English - Fluent;

Mandarin - Fluent